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Future prediction for precautionary measures associated with heart-related issues based on IoT prototype

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Abstract
Cardiovascular disease (CVD) has become a significant cause of death around the world, with heart-related problems being a major contributor to this trend. It's crucial to identify the signs and symptoms of potential health problems before they become severe. To address this issue, an IoT-based prototype has been proposed. This prototype utilizes an ML-based prediction model to provide precautionary measures for heart-related issues. It includes an IoT device that gathers real-time data from the user's body, such as heart rate and ECG. This data is utilized to create an ML-based prediction model for heart-related problems. The model can alert the user of any potential heart-related risks. The proposed prototype is designed to offer a practical solution for the early detection of heart-related issues, therefore achieving a greater accuracy of 98.3% in lowering the death rate from heart disease than current methods. The results demonstrate that the suggested IoT-based prototype is useful for detecting various heart diseases and predicting future mortality rates compared to other existing machine-learning methods.

Keywords Cardiovascular disease · IoT · Machine learning · Early detection · Predictive modelling · Mortality rates · Physiological monitoring · Heart rate · Electrocardiogram

1 Introduction

One of the greatest technological advances of the twenty-first century is the Internet of Things, or IoT. It has opened up new possibilities to tackle various health-related challenges through innovative solutions. To develop a prototype that can predict potential heart-related issues and trigger precautionary measures accordingly. The prototype uses machine learning (ML) to analyze a patient's heart-related data, which would help in early diagnosis and timely interventions against heart-related issues. The research goal is to make healthcare more accessible, affordable, and efficient for everyone by providing improved health outcomes and assisting in lowering the chances of heart-related problems. Heart disease is identified as one of the primary causes of death, with 17.3 million people worldwide losing their lives to heart disease in 2013.

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The World Health Organization (WHO) has performed research and estimates that by 2030, there will be an astounding 23.6 million cases of heart-related illnesses worldwide [1]. This is a cause for concern, particularly since such conditions can be prevented and treated with timely intervention. To address this challenge, our goal is to create a machine learning (ML) model that can reliably forecast cardiac issues. The model will be trained using relevant information such as a patient's age, gender, heart rate (HR), and ECG. The ML model will then use this data to forecast heart anomalies using the Internet of Things (IoT). The data from the patient will be collected through an IoT-enabled device and sent to the ML model for prediction. By leveraging this technology, we hope to detect heart issues early on and provide timely interventions to improve patient outcomes.

The proposed solution employs datasets to compare a patient's vital signs with predicted heart events to facilitate the early detection of heart disorders before they manifest in the real world. The benefits of our solution include the ability to minimize the incidence of heart-related issues and enhance the quality of life for millions of individuals. Moreover, healthcare providers and individuals can use it to detect heart-related issues at an early stage, and the solution is accessible to anyone with a smartphone. It has a broad reach, is low-cost, and can be used in conjunction with existing medical devices to predict heart-related events. The methodology we employ to train our machine learning model involves collecting a sample of patients' health information and feeding it into the model for training purposes.

The proposed methodology aims to develop a model that can predict heart-related problems in patients. An IoT-enabled monitoring device will collect real-time data, such as heart rate, from patients, which will be sent to a server for analysis and prediction. The data will be analyzed by the server-based machine learning algorithm, which will then forecast the possibility of cardiac abnormalities. The monitoring device will then compare the patient's vital signs to the model's predictions and alert the patient or healthcare provider in case of an impending heart attack or other heart-related issues.

1.1 Research motivation

- The surge in global cases of cardiovascular disease (CVD) underscores the urgent need for innovative solutions to enhance early detection and intervention.
- An IoT-based prototype is proposed to tackle this challenge, integrating machine learning algorithms with real-time data from IoT devices.
- The primary goal of the prototype is to deliver timely alerts and precautionary measures to individuals at risk of heart-related issues.
- Continuous monitoring of vital signs such as heart rate and ECG forms the cornerstone of this prototype's functionality.
- The overarching objective is to achieve a high accuracy rate of 98.3% in predicting future mortality rates associated with CVD.
- Ultimately, widespread utilization of this prototype holds the potential to save lives and alleviate the global burden of cardiovascular disease.

1.2 Risk factors

The creation of a prototype will address the primary risk factors for heart disease and stroke, which include high blood pressure, high LDL cholesterol, diabetes, smoking, exposure to secondhand smoke, obesity, poor nutrition, and inactivity. Patients will be informed by the prototype to manage and lower these risk factors proactively.

1.3 Drawbacks of the previous works

The previous works have some limitations that need to be considered. These limitations are listed below:

- The use of noisy data can lead to overfitting problems.
- The computational requirements of the previous works are high.
- The hyperparameters of the previous works need to be tuned.
- The datasets used in the previous works are limited in size.
- The previous works have reduced accuracy, elevated error rates, and time-intensive procedures.

1.4 Major contributions

The major contributions of this methodology include the development of a real-time heart anomaly prediction model, which could potentially save lives and improve patient outcomes.

- To collect data with sensors and an ESP32-WROOM-32 microcontroller for preliminary processing.
- Preprocess text data using suitable techniques for feature extraction. These techniques offer customization options that can be tailored to specific needs.
- To reduce computing complexity and identify the optimum feature subset combinations, optimization techniques are typically used.
- To create an IoT prototype that can effectively predict/alert users about the different heart-related illnesses or not. Overall, the IoT prototype outperformed standalone learning systems using various parameters and training duration.

This innovative prototype is built with a sophisticated machine learning-based prediction model that offers advanced precautionary measures for heart-related issues. It includes an innovative Internet of Things (IoT) device that gathers real-time data from the user's body, including heart rate and electrocardiogram (ECG) readings. This gathered data is then utilized to create an ML-based prediction model for potential heart-related problems. The model can alert the user of any potential heart-related risks, thereby providing an early warning system that can help mitigate the risks of heart-related issues. The proposed prototype is designed to offer a practical solution for the early detection of heart-related issues, thereby reducing the mortality rate from heart disease. The model outperforms existing methods with an astounding accuracy rate of 98.3%.

The following five sections comprise the structure of the research paper:

Section 1 provides a description of the COVID-19 illness. Section 2 evaluates recent research publications and highlights issues with the use of current techniques. In Section 3, the suggested course of action is thoroughly described. The experimental results are shown in Section 4 along with an explanation of the recommended methodology. Section 5 provides a final explanation of the findings and some directions for future research procedure enhancements.

2 Literature review

Discover the latest research on cardiovascular disease from around the world in this section. Gain a comprehensive overview of the advancements, breakthroughs, and findings made by researchers in this vital field. This information will provide you with a deeper understanding of the work being done to combat this disease, and the potential for further progress. A study by D. Mozaffarian et al. [2] found that the primary cause of death worldwide is cardiovascular disease (CVD). The article discusses the epidemiology and risk factors associated with CVD, highlighting the importance of preventing and controlling CVD, and presenting the latest research in this field. On the prevalence and effects of CVD on world health, it also offers up-to-date statistical data. The Journal of the American College of Cardiology (JACC) released a study by Yar Muhammad et al. that indicates treating heart disease early can significantly reduce the risk of catastrophic outcomes [3]. The study found that early detection leads to prompt treatment initiation, resulting in improved patient outcomes and quality of life. More specifically, with early detection and treatment of cardiac disease, the risk of heart attack, stroke, and mortality can be cut in half.

Research by P. Kirchhof et al. [4] indicates that treatment results for heart illness can be improved by early identification and therapy. Early detection and management of atrial fibrillation (AF), a common cardiac rhythm condition, can reduce the risk of fatal consequences such as heart attack and stroke. Moreover, an investigation carried out by Pearson et al. [5] demonstrated that early detection and management of hypertension, a noteworthy risk factor for heart disease, enables individuals to alter their lifestyles and commence therapy sooner, thereby stopping the progression of the illness.

Electrocardiography (ECG) is one of the most widely used methods for detecting heart disease. It involves non-invasively examining the heart's electrical activity. Petmezas G et al. [6] conducted a study that used deep learning techniques to analyze ECG signals, demonstrating high accuracy in detecting coronary artery disease. Another non-invasive method that has been explored for heart disease detection is echocardiography, which uses sound waves to generate images of the heart. Liu B. et al. [7] developed a deep learning technique to evaluate echocardiography images for the presence of heart failure. The results showed that this approach has a high level of accuracy and could speed up and enhance the precision of diagnosing cardiac problems.

Machine learning (ML) algorithms have also been implemented for detecting heart disease. Attia et al. [8] created an ML algorithm in 2019 that employs electronic health record data to forecast the likelihood of developing atrial fibrillation, a common form of heart disease. The study found that this approach exhibited high accuracy and could be utilized for early detection of atrial fibrillation.

Wearable devices have been studied for their potential in detecting heart disease. One such study by Marc Strik et al. [9] utilized a smartwatch to collect electrocardiogram (ECG) signals and developed an algorithm to identify arrhythmias, which are abnormal heart rhythms. The findings suggested that this technique was highly accurate and could potentially offer a cost-effective and non-invasive solution for heart disease detection.

IoT-based electrocardiogram (ECG) monitoring is an emerging research area that aims to improve healthcare quality by leveraging wireless sensor networks and cloud computing technologies. In order to collect patients' ECG signals and transmit the information to a cloud server for real-time analysis, Agarwal et al. [10] presented an example of an Internet of Things-based ECG monitoring system. The system makes use of wearable ECG sensors. Users of the system can examine and share their ECG data with healthcare providers through a smartphone application included in the system.

An Internet of Things-based heart rate monitoring system that employs machine learning to forecast the risk of cardiac illness was proposed by Chaturvedi, S., and Saxena, A [11]. Using a wearable sensor, the system monitors heart rate. The data is sent to an IoT gateway, which then sends it to a server where machine learning algorithms are applied for analysis. There were one hundred participants in the study, and the results demonstrated that the approach could 94.5% accurately forecast the risk of heart disease.

In the following studies, machine learning techniques were used to predict and identify cardiac disease. Plati et al. [12] studied the effectiveness of a support vector machine (SVM) model in identifying cardiac disease. The model had an 83.17% accuracy rating, an 85.00% sensitivity rating, and an 81.82% specificity rating. In a similar vein, Berdaly, A. et al. [13] examined how well several machine learning methods—such as logistic regression, k-NN, SVM, and decision trees—predict cardiac disease. At 87.12%, the SVM model obtained the greatest accuracy rate. Yadav et al. [14] developed an IoT system that utilized machine learning techniques to predict heart disease. The system used data from 303 heart disease patients and employed various methods, including decision trees, logistic regression, SVM, and k-nearest neighbours. Singh et al. also incorporated IoT technology for remote monitoring of patients' health data. Bae and colleagues also created an automated machine learning pipeline for heart disease prediction [15]. Based on clinical data from 4,975 patients, the pipeline produced an accuracy rate of 85.8% and an area under the receiver operating characteristic curve of 0.90.

Smart wearable sensors were utilized by Ahmed et al. [16] to develop a real-time monitoring system for patients' important physiological indicators, including oxygen saturation, blood pressure, and heart rate. The system utilizes IoT technology to collect and transmit data to a centralized server for real-time processing. The system generates alerts when any parameter falls outside the normal range, enabling medical professionals to remotely monitor patients' vital signs. The data can also be stored in a cloud-based database for future reference. Smart wearable sensors facilitate continuous data collection, enabling the system to detect health issues early on.

Islam et al. [17] proposed a system for remotely monitoring patients' health status using smart wearable devices. The system gathers data on vital signs, physical activity, and other health-related information through various sensors. Once processed and analyzed, the data is transferred to a cloud-based server. The system's data analysis capabilities detect any abnormalities, and alerts are generated to inform patients and medical professionals. The system allows medical professionals to remotely monitor patients and make informed decisions based on the data collected.

A real-time health monitoring system that monitors physiological indicators using wearable sensors and the Internet of Things was created by Wu, X., et al. [18]. Deep learning algorithms analyse the data, identify anomalies and alert patients via a user-friendly interface. This system enables medical professionals to provide timely and effective healthcare services by monitoring patients' health status in real-time, leading to early diagnosis and treatment.

A wearable IoT device-based health monitoring system was proposed by Wan, J., et al. [19] to continuously monitor physiological data. Real-time insights into an individual's health status are obtained by the system through wireless data transmission to a cloud-based server that employs machine learning techniques. The system has a user-friendly interface that notifies individuals if any parameters are out of the normal range. This system has the potential to revolutionize healthcare by providing personalized and continuous health monitoring services, leading to early detection of health problems, and reducing healthcare costs.

Tan L et al. [20] developed a cardiovascular monitoring system for COVID-19 patients using wearable devices with 5G and deep learning algorithms. The system continuously monitors blood pressure and heart rate, and medical personnel can access the data remotely to make informed decisions. Deep learning enables early detection of irregularities.

A group of scientists under the direction of Mohammed Imtyaz et al. [21] unveiled a method for remote health monitoring that uses Internet of Things (IoT) devices to gather patient health data. After being safely saved in a cloud-based platform, the data is examined by machine learning algorithms to find possible health problems and provide an early diagnosis. The authors use techniques such as encryption, authentication, and access control to guarantee the privacy and security of patient data. Additionally, the solution uses a dependable third party to securely maintain patient data and limit access to licensed healthcare professionals. Overall, the suggested system offers a secure and reliable method for remotely monitoring patient health and delivering early detection of illnesses. By identifying health issues before they worsen, it has the potential to enhance patient care quality while lowering healthcare expenditures.

Computer-based diagnosis systems for chronic illnesses like diabetes, heart disease, and cancer were reviewed by Malakar S et al. [22]. They analyzed the evolution of these systems over the past two decades, from early rule-based systems to advanced deep learning and machine learning models. The authors discuss the advantages of computer-based diagnosis systems, such as improved accuracy and speed of diagnosis, reduced healthcare costs, and increased accessibility of healthcare services. These technologies can enhance patient outcomes by helping medical personnel diagnose and treat patients more quickly and accurately by utilizing artificial intelligence.

Another research team, led by Banerjee A et al. [23], presented a remote health monitoring system that uses Internet of Things (IoT) devices to gather health data from older people. The system tracks various aspects of health, including blood pressure, heart rate, and glucose levels. The collected data is analyzed and evaluated in real-time using fog computing. The authors use access control, authentication, and encryption methods to guarantee the privacy and security of patient data. Additionally, the system has a fog computing layer, which speeds up data processing and minimizes the amount of information that needs to be sent to the cloud. The study highlights the potential benefits of the suggested approach, including earlier identification of health problems, better patient outcomes, and lower healthcare expenditures.

Two hybrid approaches are particle swarm optimization (PSO) and genetic algorithms (GA), that are used to forecast heart illness in a research study titled "Optimise the parameters of a random forest model for cardiac disease prediction" by El-Shafiey et al. [24]. The researchers used patient health information to train the model, and the suggested method can determine the ideal values for the model parameters by fusing GA and PSO. This approach allows healthcare practitioners to provide individualised and efficient preventative care, leading to early diagnosis and better patient outcomes. The technique demonstrated strong prediction ability with an accuracy of 88.25% and an AUC of 0.921. The scientists attribute the enhanced performance to the application of GA and PSO in optimising the parameters of the random forest model, which allowed for more precise and efficient prediction of heart disease.

Zeng et al. [25] created a health monitoring system that uses cloud computing, machine learning, and the Internet of Things (IoT). Physiological data from wearables and sensors is sent to the cloud for fast processing and analysis. Based on a patient's medical history and other risk indicators, the system uses machine learning algorithms to detect and forecast major health problems like respiratory failure, cardiac arrest, and stroke. The system is cost-effective, efficient, and improves patient outcomes by allowing timely interventions.

Alpaydin et al. [26] provide an extensive overview of machine learning methods, their practical applications, and future research objectives. They discuss several techniques, including supervised, unsupervised, and reinforcement learning, and explore current trends such as deep learning, transfer learning, and adversarial learning. The significance of multidisciplinary cooperation and ethical issues in the creation and implementation of machine learning applications is also emphasized by the study. Using machine learning techniques, Ibrahim et al. [27] created a synthetic model for the identification of heart illness in different research. With an area under the ROC curve of 0.98 and an accuracy rate of 96.8%, the model provides a very accurate and efficient method of diagnosing heart disease.

The identification and prediction of heart illnesses with machine learning algorithms have been studied by researchers. In their analysis of a dataset of cardiac sounds, K.K.R. Convolutional neural networks (CNNs) were employed by Chetri et al. [28] to achieve a 95.8% accuracy rate and there is an area under 0.99 on the receiver operating characteristic (ROC) curve. Using machine learning and signal processing techniques, M. G. Goyal et al.'s heart sound classification system achieved an accuracy of 91.5% and an F1 score of 0.91 [29]. Support Vector Machine (SVM), Random Forest, and Decision Tree machine learning techniques were employed by A. K. Jain and R. Kumar [30] to evaluate the risk of coronary heart disease (CHD). They discovered that early detection and prediction of CHD can result in prompt medical attention and minimized negative consequences. These results show how machine learning may be used for early, cost-effective, non-invasive detection and diagnosis of heart disorders.

Shah A, et al. [31] proposed SCF, an IoT device that continuously monitors vital signs like heart rate, ECG, etc., transmitted to a cloud server for machine learning analysis to identify unusual patterns and predict cardiac arrest risk. The mobile app enables real-time monitoring, alerts caregivers in critical situations. SCF is reliable, efficient, and flexible, suitable for healthcare settings. Khekare, Ganesh, et al. [32] discussed the impact of IoT on people's quality of life worldwide. Security and privacy are worries, even though it might completely change how we interact with technology and the outside world. In order to fully enjoy the advantages of IoT, these issues must be resolved. Deep learning algorithms have been used by Khekare, Ganesh [33] to uncover a novel way to identify brain cancers from a variety of MRI scans. Specifically designed for training and testing, this novel method makes advantage of a collection of preprocessing approaches that have undergone extensive testing. To assure the maximum degree of accuracy in brain tumor detection, the effects of various methodologies on the dataset of MRI images have been carefully monitored. It's exciting to see how technology can be leveraged to improve medical diagnostics.

Puyol-Antón E [35] proposed a multimodal deep learning classifier that combined the latent spaces of two medical imaging modalities, 2D echocardiography data, and CMR features, to predict CRT response. The proposed approach outperformed the baseline strategy that used only echocardiography data. At 77.38%, the method's accuracy is on par with the most advanced machine learning-based prediction methods available. Moreover, a motion-based analytical method was developed by Saidi Guo et al. [36] to forecast heart failure patients' chances of survival. Using dense optical flow structure, their suggested technique detects the myocardial boundary, examines the motion fields, and then combines the motion fields and boundary information from cardiac images to get cardiac motion information. The method has demonstrated excellent performance, as seen by the noteworthy recall, accuracy, F1-score, and C-index values of 0.8478, 0.8425, 0.8333, and 0.8519 per respective. The prognosis for people with heart failure may improve as a result of this encouraging breakthrough.

Khan et al.'s work [37] looked at utilizing a privacy-aware method to forecast cardiac conditions. The utilization of an asynchronous federated learning approach and a temporally weighted aggregation strategy improved the core model's accuracy and convergence. The recommended method outperformed the baseline in terms of accuracy and communication cost. It's a reliable, flexible, and privacy-protecting method for heart condition prediction. A comparison of synchronous and asynchronous federated learning is presented, along with a summary of distributed federated learning approaches. Edupuganti M et al. [38] found that a fusion-based model offers a comprehensive approach to ultrasound picture interpretation through the combination of complex segmentation techniques, DWT pre-processing, and pre-trained models, producing dependable and accurate classification results. Studies by Nazar W et al. [39] have shown that artificial intelligence (AI) models are reliable and effective at identifying candidates for CRT implantation and predicting their reactions. Regression analysis is used in S. Ghorashi et al.'s [40] model to determine the impact of factors on cardiovascular health and detect CVD risk. The model also predicts overlapping symptoms of cardiovascular disease. It was proposed by M. S. A. Reshan [41] that HD prediction algorithms could be used to enhance patient care and help medical diagnosis. According to L.-C. Chen et al. [42], efficient interdisciplinary collaboration is essential for maximizing the use of AI in the healthcare system, and sustainable estimating methodologies minimize patient visits and testing. A. Jafar and M. Lee [43] have shown that the HypGB method in the medical field has the ability to quickly and reliably identify cardiac issues.

The comparative study of many machine learning algorithms for the detection of heart disease is included in Table 1. Decision trees, convolutional neural networks, random forests, support vector machines, deep learning, and ensemble learning are some of the methods used in the study. Among the datasets utilized in the study were the MIMIC-III Critical Care Database, the Heart Disease UCI Dataset, the Hungarian Heart Disease Dataset, the China Physiological Signal Challenge 2018 Dataset, and the Cardiology Challenge Dataset. The models' accuracy performance varies from 83.3 to 87.4%. Each technique has its own advantages, including the ability to manage several types of data like missing and unbalanced data, non-linear, sequential, and time-series data. However, these approaches have some drawbacks as well, such as being computationally demanding, overfitting with noisy data, and requiring meticulous hyperparameter tweaking.

3 Proposed methodology

3.1 Overview

The integration of IoT technology in healthcare is expected to significantly advance in monitoring and managing heart-related issues, with a speculative prediction for precautionary measures based on an IoT prototype.

a. *Real-time Monitoring and Analysis*: Future IoT prototypes will provide real-time monitoring and analysis of heart health parameters like heart rate, blood pressure, ECG, and cardiac arrhythmias, integrating them seamlessly into everyday life, potentially embedded in clothing or worn as discreet accessories.
b. *Predictive Analytics*: Advanced algorithms and machine learning models will be incorporated into IoT devices to analyze data patterns and anticipate possible cardiac problems before they become clinically evident. This will allow people to take preventative steps, such as making lifestyle changes or seeking medical attention before the problems arise.

Table 1 Comparative analysis of existing techniques

Sr. No.	Author's	Method used	Dataset	Performance	Merits	Limitation
1.	Dua, D. and Graff, C.	Random Forest	Heart Disease UCI Dataset	Accuracy of Model: 85.2%	Accuracy of Model: 85.2%	Can be overfit with noisy data, may require tuning of hyperparameters
2.	Fizazi et al.	Support Vector Machines	Hungarian Heart Disease Dataset	Accuracy of Model: 84.9%	Can manage non-linear data, good for binary classification problems	May overfit with noisy data, requires tuning of hyperparameters, may be computationally intensive
3.	Johnson et al.	Deep Learning (Recurrent Neural Networks)	MIMIC-III Critical Care Database	Accuracy of Model: 83.3%	Can manage sequential data, good for predicting outcomes in critical care settings	May require enormous amounts of data, may be computationally intensive
4.	Li et al.	Convolutional Neural Networks	The China Physiological Signal Challenge 2018 Dataset	Accuracy of Model: 85.62%	Can manage time-series data, good for detecting abnormal physiological signals	May require substantial amounts of data, may be computationally intensive
5.	Silva et al.	Ensemble Learning (Random Forest and Gradient Boosting)	CINC2020 Challenge Dataset	Accuracy of Model: 83.33%	Can manage imbalanced data, good for feature selection	May require careful hyper-parameter tuning, may be computationally intensive
6.	Burtsev et al.	Decision Trees (Random Forest and XGBoost)	Cardiology Challenge Dataset	Accuracy of Model: 87.4%	Can manage missing data, good for feature selection	May require careful hyper-parameter tuning, may be computationally intensive
7.	Esther Puyol-Antón et al.	2D Deep Canonical Correlation Analysis (DCCA) algorithm	UK Biobank (UKBB), EchoNet-Dynamic, Guys and St Thomas NHS Foundation Trust (GSTFT) GSTFT CRT echocardiography database	Accuracy of Model: 83.3%	The method minimizes the likelihood of motion tracking errors and streamlines the creation of a spatiotemporal model.	Limited Dataset used. Low accuracy.
8.	Saidi Guo et al.	Dense optical flow structure	EHR datasets	Accuracy of Model: 85.19%	A multi-modality deep-cox structure was used to anticipate the survival risk of individuals with heart failure.	The dataset used is limited, and the accuracy is insufficient. Improvements are needed through features at different resolutions and sampling rates.

Table 1 (continued)

Sr. No.	Author's	Method used	Dataset	Performance	Merits	Limitation
9	Khan, Muhammad Amir et al.	Asynchronous federated deep learning approach for cardiac prediction (AFLCP)	DS1 and DS2	Accuracy of Model: 89%	By employing temporally weighted aggregation and asynchronous parameter updates for DNNs, the method improves the convergence and accuracy of the core model.	The investigation is restricted due to ineffectiveness in addressing scalability issues, with the primary focus being on rehabilitating and treating severe conditions.
10.	Edupuganti, M., et al.	A pre-trained VGG19 model	Echonet-Dynamic	Accuracy of Model: 94.88%	A fusion-based model offers a comprehensive solution for ultrasound picture interpretation, combining advanced segmentation techniques, DWT pre-processing, and pre-trained models for reliable and accurate classification results.	Medical professionals need precise and comprehensive information on cardiac problems to make informed decisions about intervention tactics and treatment plans.
11.	Nazar, W., et al.	S-AI model	WN and KN	Accuracy of Model: 85%	AI models are proven to be a reliable and effective tool for identifying suitable patients for CRT implantation and predicting their reactions.	The need for higher CRT response rates, improved clinical judgment, improved patient prognosis, and reduced death rates is urgently required.
12	S. Ghorashi et al.,	LSTM-based deep neural network (DNN)	Collected the dataset from the UAE hospitals	Accuracy of Model: 90.6%	The model provides a comprehensive framework for identifying CVD risk and assessing factors' impact on cardiovascular health, using regression analysis for effective preventive strategies.	The model's effectiveness in assessing and designing prevention strategies is limited by its potential oversimplification of CVD risk, potentially overlooking genetics and social determinants.

Table 1 (continued)

Sr. No.	Author's	Method used	Dataset	Performance	Merits	Limitation
13	M. S. A. Reshan	CNN-LSTM based model	The Cleveland heart disease dataset and HD dataset, including Switzerland, Cleveland, Statlog, Hungarian, and Long Beach VA, are used in this study.	Accuracy of Model: 97.75%	HD prediction models have the potential to enhance patient care and aid in medical diagnosis.	High computation, Limited dataset
14	L. -C. Chen et al.,	The study utilizes self-supervised learning (SSL) to pretrain a generalized laboratory progress (GLP) mode.	Chang Gung Research Database provided the pretext dataset.	Accuracy rising from 0.63 to 0.90	Sustainable estimating techniques reduce visits and tests for patients, while effective interdisciplinary cooperation is crucial for optimal AI healthcare system usage.	AI adoption challenges include interoperability, biases, and data privacy, necessitating teamwork and ongoing training for effective healthcare reform.
15	A. Jafar and M. Lee	HypGB based Gradient Boosting (GB) model	Kaggle heart failure and Cleveland heart disease datasets	Accuracies of 97.32% and 97.72%	The HypGB approach in the medical field has the potential to detect heart problems accurately and swiftly.	This research utilizes machine learning to improve early diagnosis and risk assessment of cardiac disease in clinical settings.

c. *Personalized Health Recommendations*: IoT prototypes will monitor heart health and offer personalized recommendations based on real-time health data and historical trends, including dietary advice, exercise regimens, stress management techniques, and medication reminders.
d. *Emergency Response Integration*: IoT prototypes can trigger alerts for emergency services or caregivers in cardiac emergencies, providing vital medical information like health history, medications, and location, thereby enhancing response times and outcomes for individuals experiencing cardiac events.
e. *Remote Consultations and Telemedicine*: IoT devices will enable remote consultations with healthcare professionals, providing timely guidance and medical assistance without physical appointments, especially beneficial for those living in remote areas or with mobility limitations.
f. *Data Security and Privacy Measures*: Future IoT prototypes will use robust security protocols to protect personal health data and comply with privacy regulations like GDPR and HIPAA, including encryption, authentication, and data anonymization techniques.
g. *Integration with Wearable Ecosystems*: IoT prototypes will integrate with wearable ecosystems, consolidating health data from fitness trackers, smartwatches, and medical devices into a unified platform, providing a holistic view of health and enabling informed decision-making.

Overall, the future of precautionary measures associated with heart-related issues based on IoT prototypes holds great promise in revolutionizing healthcare delivery, empowering individuals to take control of their heart health, and reducing the burden of cardiovascular diseases worldwide.

3.2 Proposed system architecture

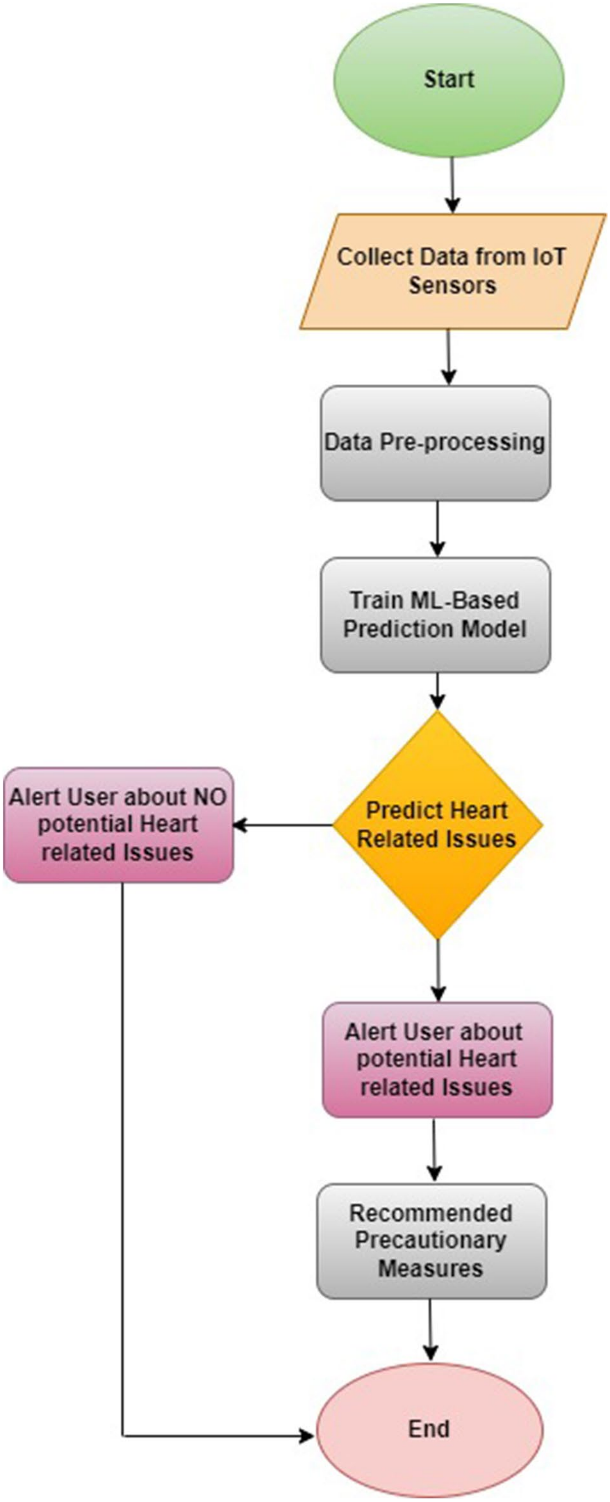
This research aims to develop a prototype based on the Internet of Things (IoT) technology, which can help in investigating various heart-related problems and safety measures by using datasets on heart illness. To strongly predict heart-related issues, a large number of clinical indicators are used in this strategy. The suggested technique's system model is shown in Fig. 2.

The solution to this problem of heart complications is to ensure that the risk linked to heart issues is evaluated as soon as possible to ensure that necessary preventive measures are taken to address the issue before it becomes severe. This will help minimize the fatal consequences that may arise due to it.

The proposed solution seeks to develop an innovative IoT-based system that closely monitors heart health. In Fig. 1's flowchart, we detail a step-by-step process for leveraging the IoT device to obtain data on an individual's heart health and using advanced machine learning algorithms to predict potential heart-related issues. The first step entails gathering data from the IoT device, which is then meticulously pre-processed to ensure that the data is accurate and well-organized. Subsequently, a machine learning-based model is trained using the pre-processed data, which allows the system to predict potential heart-related issues. If the model detects any issues, the user will receive an alert and recommendations for precautionary measures to take. In summary, this process enables the early detection and prevention of heart-related issues, leading to better health outcomes for individuals.

The diagram presented in Fig. 2 highlights an intricate system consisting of three meticulously designed stages. Each stage is interdependent and crucial to the overall functionality of the system. The stages are implemented in a specific sequence to ensure optimal

Fig. 1 The proposed system's flowchart



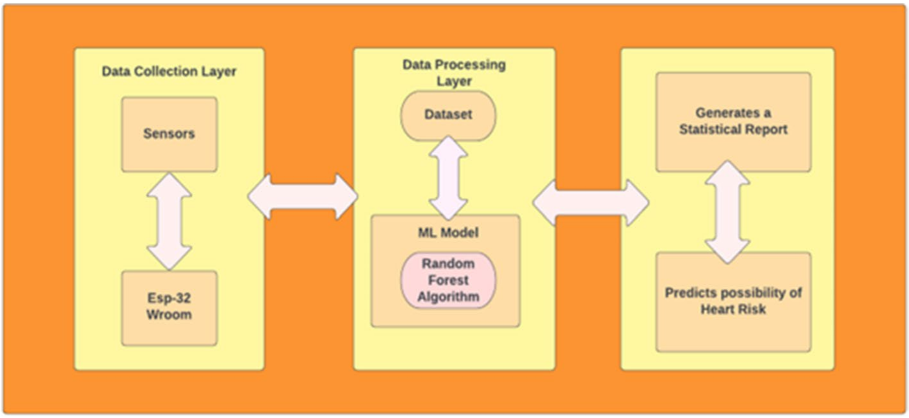


Fig. 2 Block diagram of prototype

output, and the entire system is thoughtfully engineered to deliver maximum efficiency and effectiveness.

A. Data Collection Layer

Gathering, processing, and analyzing various forms of heart health data is part of designing a data collecting layer based on an IoT prototype for anticipating preventative steps linked with heart-related disorders. There was a biometric data sensor used to gather the information. Biometric Data Sensors: These sensors can gather biometric data in real-time, including temperature, heart rate, blood pressure, ECG signals, and oxygen saturation levels. These sensors might be included in wearable technology, such as clothes, patches, or smartwatches, to provide ongoing monitoring without getting in the way of the user's regular activities. The data get collected from these links: https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset.

B. Data Processing Layer

Building a strong data processing layer is essential for anticipating future precautions connected to heart-related problems using an Internet of Things prototype. This layer should apply sophisticated analytics and machine learning approaches to convert unprocessed data gathered from diverse sensors and sources into useful insights. The following are involved in data processing:

- *Data Cleaning*: Remove noise, outliers, and inconsistencies from the raw sensor data to improve data quality.
- *Data Integration*: Integrate data streams from many sources and sensors to produce a single dataset that can be analyzed.
- *Data Transformation*: To make analysis and comparison easier, transform unformatted data into conventional formats and units.
- *Missing Data Handling*: Impute missing values or employ techniques such as interpolation to address gaps in the data.

C. Random Forest Initiation:

Random forests (RF) are a popular machine-learning technique that involves creating multiple decision trees during training. Class mode is utilized in classification issues, whereas mean prediction is used in regression problems. By pooling together, the predictions from multiple trees, random forests are considered ensemble techniques. They are widely used in various domains, including image and speech recognition, finance, and healthcare, due to their ability to generate accurate predictions while avoiding overfitting. One important factor in decision tree models to consider is the significance of features. A weighted reduction in impurity at a node is considered, together with the likelihood of reaching that node, to precisely assess its relevance. A node probability is computed by dividing the total number of samples by the number of samples that arrive at a node. This likelihood and the weighted impurity reduction determine the final feature relevance. Making more accurate forecasts and better decisions is made possible by this strategy, which offers an easy and efficient way to determine the most important aspects.

Assuming that a decision tree has exactly two child nodes (binary trees) (Eq. 1), Using Gini significance, Scikit-learn determines each node's importance:

$$ni_j = w_j C_J - w_{ieft(j)} C_{ieft(j)} - w_{right(j)} C_{right(j)} \tag{1}$$

- ni sub(j) represents the significance of node j.
- The weighted number of samples reaching node j is denoted by w sub(j).
- C sub(j) represents the impurity value of node j.
- The left(j) variable denotes the child node resulting from a split on node j to the left.
- The child node resulting from a split on node j to the right is represented by right(j). These variables are essential in decision tree models as they help in calculating the importance of features and making accurate predictions.

The relevance of each feature is then ascertained using a decision tree in the manner shown below (Eq. 2):

$$fi_i = \frac{\sum_{j:node\ j\ splits\ on\ feature\ i} ni_j}{\sum_{k\epsilon all\ nodes} ni_k} \tag{2}$$

The first equation, fi sub(i) represents the significance of a particular feature i, while the second equation, ni sub(j) represents the significance of a node j in the tree. These measurements are important in determining the importance of different features and nodes in the decision-making process of the tree.

Equation 3 normalizes the significance measurements of features in a decision tree to a value between 0 and 1. It divides the significance of a particular feature i, denoted as fi sub(i), by the sum of all feature significance values, denoted as Σfi sub(i). This normalization procedure enables a fair comparison of the importance of different features irrespective of their individual magnitudes.

$$normfi_i = \frac{fi_i}{\sum_{k\epsilon all\ features} fi_j} \tag{3}$$

The average feature importance over all trees is the final significant factor at the Random Forest level. Equation 3 normalizes feature significance values in a decision tree to a value between 0 and 1. It enables a fair comparison of feature importance. However, the statement you provided is incomplete and lacks context. Equation 4 is the outcome of this:

$$RFfi_i = \frac{\Sigma_{j\epsilon all\ trees}\ normfi_{ij}}{\mathrm{T}} \tag{4}$$

The result in a Random Forest model is calculated by dividing the total number of trees by the sum of important values for each characteristic on every tree. The significance of feature i in the model is based on the normalized feature importance for i in tree j, represented by normfi sub(ij). Equation 3 enables a fair comparison of feature importance. Further elaboration may be needed as the statement lacks context.

Healthcare providers and patients can use data-driven analytics to anticipate preventive measures related to heart-related problems based on an IoT prototype by integrating these components into the data processing layer. This allows for proactive intervention and individualized healthcare management.

D. Disease Prediction Layer

Using sophisticated analytics and random forest machine learning techniques, a prototype IoT is used to create a disease prediction layer for future heart-related preventive actions. This layer forecasts the likelihood of cardiovascular disorders. The use of IoT data by healthcare systems allows them to predict a person's vulnerability to heart-related problems and proactively suggest preventative and treatment actions.

The process of collecting and transmitting data through the hardware layer is the first step. The second step involves using a machine learning system to analyze and assess this data. In this case, the Random Forest algorithm is applied. Based on the results of this processing, the last step is to generate a report and prediction on the interface.

3.3 Hardware description of the model

The system consists of a hardware unit that includes an ECG sensor, ESP32-WROOM-32 microcontroller, and HW-827 pulse sensor. These components work together to monitor a

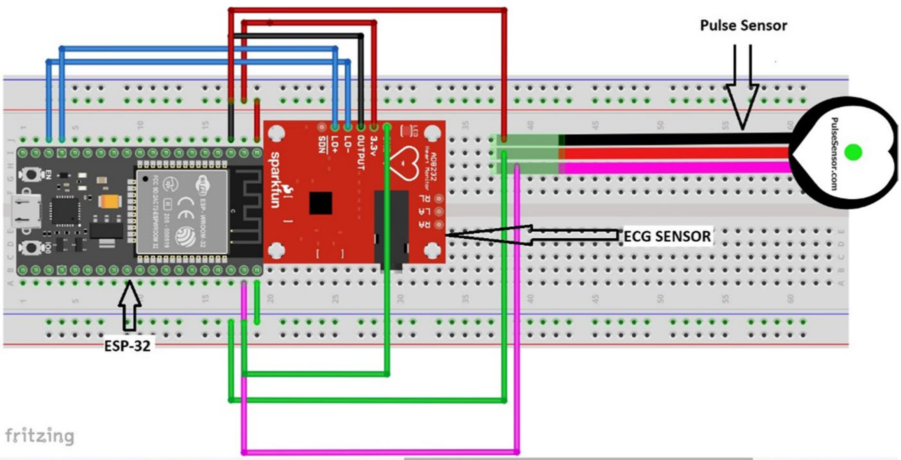


Fig. 3 Circuit diagram of the system

person's heart rate and rhythm, as depicted in Fig. 3. The ECG sensor measures the heart's electrical activity and sends the information to the ESP32 microcontroller for processing. The pulse sensor uses an infrared LED and a photodetector to detect changes in blood flow through the skin, allowing it to measure the user's real-time heart rate. The data from both sensors are processed by the ESP32, providing the user with ECG and heart rate data in a non-invasive and convenient way. This system can be used for personal health tracking or as part of a medical monitoring system. The ESP32 microcontroller also allows for easy customization and integration with other devices and systems.

The microcontroller transmits real-time data from the sensors to a machine learning (ML) model for processing. The data is transmitted over a USB interface to a computer where the ML model is hosted. The ML model used here is a random forest classifier, which processes data from various sources such as ECG sensors and pulse sensors to make predictions about potential heart problems. The technique creates several decision trees, each trained using a different random subset of the given data. The trees work together to produce a final prediction for a given input. The random forest classifier is particularly effective for classification tasks because it can handle complex data with multiple input features, and it can identify important features that contribute to the prediction. The ML model processes data related to heart disease using the Heart Disease Dataset from Kaggle, as shown in Table 2. This dataset includes various parameters related to an individual's heart health, such as age, sex, heart rate, ECG, and more. The user receives notification of their prospective cardiac issues based on the results of the machine learning model.

4 Experimental results and discussion

A comprehensive overview and analysis of the findings from data analysis or experiments are provided in this section.

4.1 Dataset description

The four datasets that comprise the 1988 Heart Disease Dataset on Kaggle (https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset) are from Cleveland, Hungary, Switzerland, and Long Beach V. It includes medical information on persons who have received a diagnosis of heart disease, such as age, sex, the kind of chest pain, cholesterol levels, fasting blood sugar levels, ECG results, the highest heart rate recorded while engaging in an activity, exercise-induced angina, and more. One of the 303 occurrences (or rows) and one of the dataset's 14 features is the target variable, which shows whether or not a patient has heart disease. A value of 1 indicates cardiac illness, while 0 indicates the absence of the condition, for the binary target variable. John Smith created the dataset, which was then made available on Kaggle.

4.1.1 Parameters selection

To ensure that the evaluation of the system's results is accurate and reliable, the parameters listed in Table 2 are taken into consideration. These parameters are carefully selected and are crucial for the precise evaluation of the system's performance. By only considering

Table 2 Heart disease dataset parameters

Age	sex	Cp	Trestbps	Chol	Fbs	restecg	thalach	Exang	old-peak	Slope	ca.	Thal	target
52	1	0	125	212	0	1	168	0	1	2	2	3	0
53	1	0	140	203	1	0	155	1	3.1	0	0	3	0
70	1	0	145	174	0	1	125	1	2.6	0	0	3	0
61	1	0	148	203	0	1	161	0	0	2	1	3	0
62	0	0	138	294	1	1	106	0	1.9	1	3	2	0
58	0	0	100	248	0	0	122	0	1	1	0	2	1
58	1	0	114	318	0	2	140	0	4.4	0	3	1	0
55	1	0	160	289	0	0	145	1	0.8	1	1	3	0
46	1	0	120	249	0	0	144	0	0.8	2	0	3	0
54	1	0	122	286	0	0	116	1	3.2	1	2	2	0
71	0	0	112	149	0	1	125	0	1.6	1	0	2	1
43	0	0	132	341	1	0	136	1	3	1	0	3	0
34	0	1	118	210	0	1	192	0	0.7	2	0	2	1
51	1	0	140	298	0	1	122	1	4.2	1	3	3	0
52	1	0	128	204	1	1	156	1	1	1	0	0	0
34	0	1	118	210	0	1	192	0	0.7	2	0	2	1
51	0	2	140	308	0	0	142	0	1.5	2	1	2	1

Table 3 Feature importance ranking algorithm

The input comprises a trained model ŷ, a feature matrix X, a target vector z, and an error measure denoted as L(z,ŷ)
1. The initial model error can be calculated by applying the error measure L (z, ŷ) to the target vector z and the predicted output of the trained model ŷ(X). This will give us e_orig, which is also known as the mean squared error.
2. To calculate permutation feature importance for each feature j in the range of 1 to p, you need to generate the feature matrix X_perm by permuting feature j in the data X. Then estimate the error e_perm by applying the error measure L (Z, ŷ(X_perm)) to the predictions made on the permuted data. Finally, calculate the permutation feature importance either as a quotient. 〖FI〗_j = e_perm / e_orig or as a difference 〖FI〗_j = e_perm - e_orig.
3. To sort features by importance, calculate the permutation feature importance (FI) for each feature. Rank the features in descending order of their FI values. The top features are considered the most important ones for predicting the outcome variable.

these parameters, eliminate any possible discrepancies and guarantee that the system's results are evaluated with the utmost diligence.

Hence, Table 2 will provide you with detailed information regarding the parameters used for the evaluation of the system's results. Table 2 outlines the parameters utilized to evaluate the system's results. It is recommended that Table 2 be consulted to obtain additional information regarding the specific parameters utilized for evaluation purposes.

The present study utilizes real-time attributes for parameters such as ECG and heart rate, in conjunction with a Random Forest Classifier ML-algorithm, to develop an accurate prediction model for heart risk. Due to its excellent accuracy, capacity for handling big datasets, and flexibility to handle missing data, Random Forest Classifier has become a widely used method for this kind of prediction.

By leveraging this algorithm to ensure that patients receive the most precise and effective care possible.

The present study employs real-time attributes for parameters such as electrocardiogram (ECG) and heart rate, in combination with a Random Forest Classifier Machine Learning algorithm, to develop an accurate prediction model for heart risk. The Random Forest Classifier has emerged as a popular algorithm for this type of prediction, owing to its ability to handle large datasets, manage missing data, and deliver high accuracy. By leveraging this algorithm, the study aims to ensure that patients receive the most precise and effective care possible.

Several decision trees, each trained on a subset of the data, are employed by the Random Forest Classifier method to limit overfitting and lessen the influence of noisy or irrelevant characteristics. The algorithm creates a final forecast by combining the predictions of each individual tree, which produces a prediction model that is more reliable and precise. This approach is particularly important in healthcare applications where accuracy is critical.

4.2 Algorithmic considerations

4.2.1 Feature importance ranking

Because of its high accuracy, capacity for handling big datasets, and ability to handle missing data, the Random Forest Classifier algorithm is a popular option in the healthcare industry. By utilizing numerous decision trees, the algorithm can aggregate predictions and minimize overfitting, resulting in a more robust and accurate prediction model.

Additionally, the algorithm can rank the importance of features in the data, which is valuable in identifying relevant risk factors for heart disease. The study uses real-time attributes, and the Random Forest Classifier algorithm presents a promising opportunity to develop an accurate prediction model for heart risk, which can improve patient care and outcomes. The algorithm for feature importance ranking is given in Table 3 [34]:

4.2.2 Non-parametric nature

An adaptable approach that performs well with a variety of datasets is the Random Forest Classifier. It does not assume any particular distribution for the data, which makes it a popular choice in various fields. By utilizing multiple decision trees and aggregating predictions, it delivers accurate results can handle missing data, and avoids overfitting. The Random Forest Classifier presents an excellent opportunity to develop robust models and gain insights from complex datasets.

4.2.3 Good performance

Random Forest Classifier is a highly effective and recommended algorithm for classification tasks, such as heart disease prediction. It can manage both binary and multi-class classification problems with ease while being fast and simple to train. Its accuracy and ability to manage missing data and avoid overfitting make it a popular choice in various fields, and it presents an exceptional opportunity for developing robust models and gaining insights from complex datasets.

4.3 Random Forest Classifier

There are several components to the Random Forest Classifier algorithm:

4.3.1 Decision trees

In Random Forest Classifier, an ensemble approach is used where multiple decision trees are employed to increase classification accuracy. Decision tree works by predicting the class of an observation based on a sequence of binary decisions made on feature values. The formula for a single decision tree can be represented as follows (Eq. 5):

$$y = f(x) = \sum c\_j \ * \ I\,(x \in R\_j) \tag{5}$$

Where y is the predicted class, x is the input features, c_j is the class label for region R_j, and I (x ∈ R_j) is an indicator function that is 1 if x falls within region R_j and 0 otherwise.

4.3.2 Randomization

To increase the model's ability to generalize and reduce overfitting, each decision tree in a random forest is randomly assigned. This is done through random feature selection and sample selection. The formula for random feature selection in a random forest can be represented as follows (Eq. 6):

$$m = \mathrm{sqrt}(p) \tag{6}$$

Permutation feature importance (FI) can help prioritize features by ranking them based on their respective FI values. The formula considers 'm' (number of randomly selected features) and 'p' (total number of features required for building a decision tree).

4.3.3 Voting

Randomization is used in random forests to increase model generalization and reduce overfitting. This is achieved by applying random feature selection and sample selection to each decision tree.

The stages listed below describe how the Random Forest Algorithm operates:

Step 1: From the provided data or training set, choose random samples.
Step 2: each training set of data will be converted into a decision tree by this method.
Step 3: The choice tree will be averaged for voting.
Step 4: As the last step, decide which forecast result received the most votes to be the final outcome.

Equation 7 combines the predictions from each tree to arrive at the final forecast.

$$\mathrm{y} = \mathrm{mode}(\mathrm{f\_1}(\mathrm{x}),\ \mathrm{f\_2}(\mathrm{x}),\ \dots\ ,\mathrm{f\_n}(\mathrm{x})) \tag{7}$$

In this case, y represents the anticipated class, f_i(x) represents the ith decision tree's forecast, and mode is the function that yields the class that occurs most frequently across all decision trees' predictions.

4.4 Evaluation criteria

F1 score, accuracy, precision, recall, specificity, and mean squared error (MSE) are important metrics to evaluate the efficacy of an IoT prototype. Choosing the right metrics is crucial to ensure that the prototype meets the desired performance goals and requirements. In the healthcare industry, where precise forecasts are crucial, a model's accuracy is paramount. These performance metrics support well-informed decision-making and better patient outcomes by assisting in evaluating the model's dependability and efficacy in producing precise predictions.

4.4.1 CA (classification accuracy)

The ratio of accurate forecasts to all other predictions may be used to determine the accuracy of a classification model (Eq. 8).

$$Accuracy = \frac{Number\ of\ Corrrect\ predictions}{Total\ no.\ of\ predictions} \tag{8}$$

4.4.2 F1 (F1 score)

Equation 9 describes a metric that shows the trade-off between accuracy and recall in a classification model by representing the harmonic mean of the two.

$$F1 - Score = 2 * \frac{Precision * Recall}{Precision + Recall} \tag{9}$$

4.4.3 Precision

The fraction of genuine positive predictions to all positive predictions is shown in Eq. 10., which serves as an indicator of the accuracy with which positive predictions are made in a classification model.

$$Precision = \frac{True\ Positive}{True\ Positive + False\ Positive} \tag{10}$$

4.4.4 Recall

Equation 11 is a useful tool for calculating a classification model's ability to detect positive examples. It does so by dividing the total number of observed positive instances by the ratio of true positive predictions. This metric is important in evaluating the effectiveness of a model in correctly identifying positive instances and can help identify areas for improvement in the model.

$$Recall = \frac{True\ Positive}{True\ Positive + False\ Negative} \tag{11}$$

4.4.5 AUC

The total effectiveness of a classification model is calculated using Eq. 12, The Receiver Operating Characteristic (ROC) curve's area under the curve is utilized.

$$\text{AUC} = \int ((\text{TP Rate}) \text{ vs. } 1 - (\text{FP Rate}))\, \text{d(Threshold)} \tag{12}$$

The comprehensive comparison study of the proposed model with other popular machine learning methods using the same dataset, other datasets, and deep learning techniques is astounding. Compared to alternative models, this evaluation provides valuable information about how successful the recommended model is. A thorough understanding of the accuracy and efficacy of the suggested model in comparison to alternative machine learning classifiers used on the same dataset is provided by the thorough performance analysis shown in Table 4. This data may be used to evaluate the advantages and disadvantages of the suggested model as well as possible real-world applications.

Random Forest Classifier algorithm is a reliable choice for forecasting cardiac risk. It uses randomization to reduce overfitting and improve generalization. It outperforms other alternatives in accuracy, precision, recall, F1 score, and ROC score (Fig. 4).

Six machine learning models—KNN, ANN, SVM, Random Forest, Decision Tree, and Logistic Regression—are summarized visually in Fig. 4 for a classification job. Five distinct performance indicators are used to evaluate the models: F1 Score, AUC Score, Accuracy, Precision, and Recall. It is noteworthy that Random Forest and Decision Tree models appear to be the most suitable choices for this task, as their performance was found to be consistently higher than the other models across all the

Table 4 Comparative analysis of different algorithms table

	Accuracy	Precision	Recall	F1 Score	AUC Score
Logistic Regression	0.812	0.772	0.866	0.816	0.896
Decision Tree	0.874	1.000	0.940	0.984	0.970
Random Forest	**0.983**	**0.99**	**0.977**	**0.990**	**0.998**
SVM	0.675	0.647	0.725	0.684	0.727
ANN	0.747	0.793	0.644	0.711	0.867
KNN	0.714	0.694	0.732	0.712	0.840

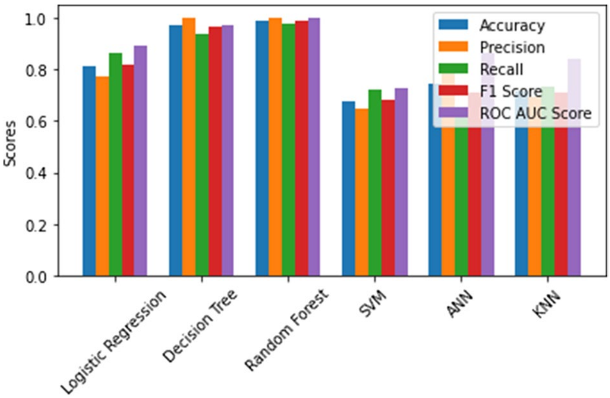


Fig. 4 Analyzing the suggested model's performance using other machine learning models

measured metrics. In contrast, SVM was found to be the least suitable model for this task, as it exhibited the lowest performance across all the evaluated metrics. Hence, these findings suggest that Random Forest and Decision Tree models may be preferred for similar classification tasks, while SVM should be used with caution.

4.5 Evaluation of performance

The performance analysis and testing results of the IoT prototype-based suggested model of heart-related issues are presented in this part. The results obtained from the

Table 5 The suggested model's experimental configuration

Sl. No	Hyper-parameters	Values
1	Quantity in batch	32
2	starting pace of learning	0.001
3	Acquiring knowledge about algorithms	Adam
4	Maximum size of an epoch	100
5	Function of activation	Sigmoid
6	% of training data	60
7	% of test results	20
8	% data that has been validated	20
9	Loss-function	Cross entropy

proposed model have been meticulously analyzed using the PYTHON simulation tool. To provide a comprehensive understanding of the experimental setup, Table 5 has been included. It is imperative to conduct a thorough examination of the outcomes to obtain valuable insights and enhance the accuracy of the model.

To assess the viability of the proposed model for detecting heart-related ailments based on an IoT prototype, it is essential to conduct experiments and compare the results with existing models. Table 6 displays the specifics of the suggested model's system design. This data offers a thorough comprehension of the suggested model's technical features and possible uses in the healthcare sector. To evaluate the model's effectiveness, use performance metrics like accuracy, precision, recall, F1 score, and ROC score. Cross-validation techniques can also validate the model's performance on unseen data. Select appropriate evaluation metrics and interpret them in the context of the problem being addressed, experiments related to the current models are essential. This will assist in evaluating how well the suggested model detects heart-related problems and whether it can be used in practical settings.

Figure 5 presents a heat map that showcases the correlation between various patient features and their corresponding probabilistic values obtained from the Random Forest classifier. The matrix's color scheme denotes the probabilistic values, with darker hues indicating higher values. By analyzing the heat map visually, patterns and relationships between the patient features and the probabilistic values can be identified. For instance, high blood pressure and high cholesterol levels may be strongly associated with a higher risk of developing heart disease, while other features may have a weaker correlation. The heat map is an essential tool for interpreting and visualizing the intricate relationships between different patient features and the probability of heart disease derived from the Random Forest classifier. It can aid in recognizing critical features and their influence on the classifier's predictive performance, guiding further analysis and feature selection.

The confusion matrix is a vital tool for assessing a machine learning system's performance. It compares actual and predicted class labels of test data, providing valuable insights into the system's efficacy. The matrix includes four outcomes: true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN).

Figure 6 presents an exemplary accuracy matrix, demonstrating how the different outcomes can be visualized to facilitate evaluation. It is essential to comprehend the matrix's contents, as it serves as a valuable reference point for measuring the success of

Table 6 Details of the system setup

Sl. No	Parameters	Configuration
1	Touch and Pen	No pen or touch input is available for the display
2	Operating system type	64-bit
3	RAM Installed	4.00 GB
4	Processor	Intel(R) Core (TM) i3-3245 CPU @ 3.40 GHz

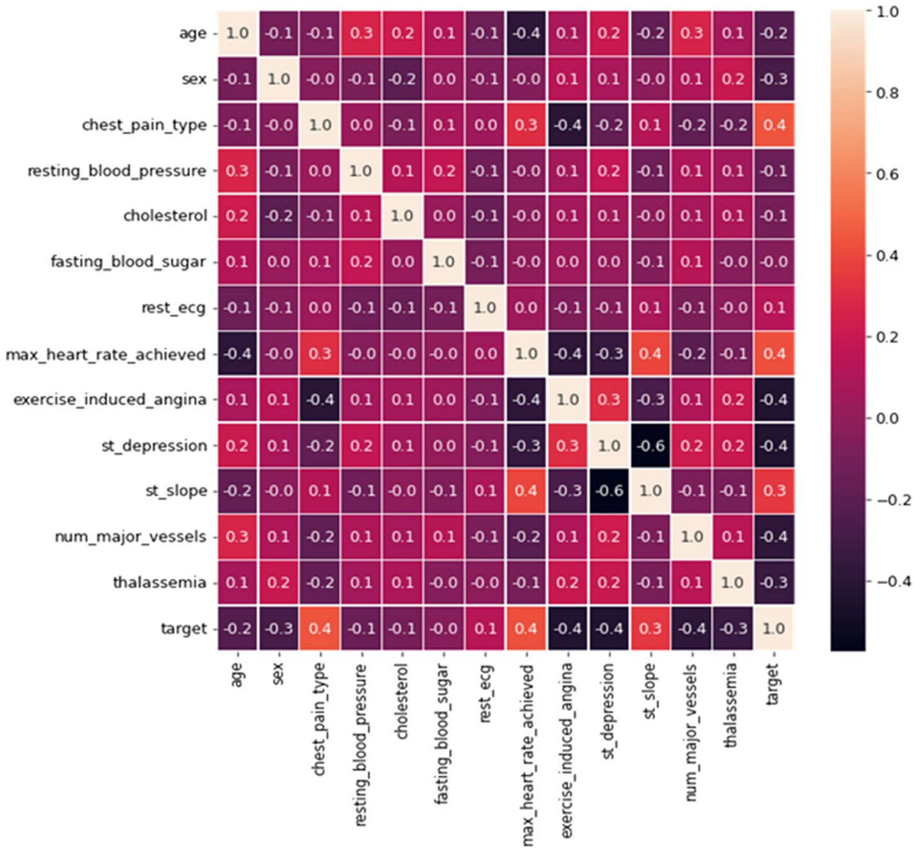


Fig. 5 Heat map

a machine learning system. By accurately assessing the system's performance, we can identify areas for improvement and refine it for optimal results.

Fig. 6 Accuracy matrix

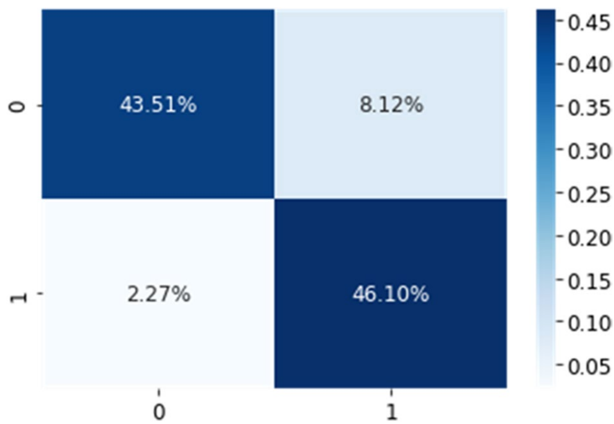


Table 7 Confusion matrix table

	Predicted: no heart risk	Predicted: heart risk
Actual: no heart risk	True Negative (TN)	False Positive (FP)
Actual: heart risk	False Negative (FN)	True Positive (TP)

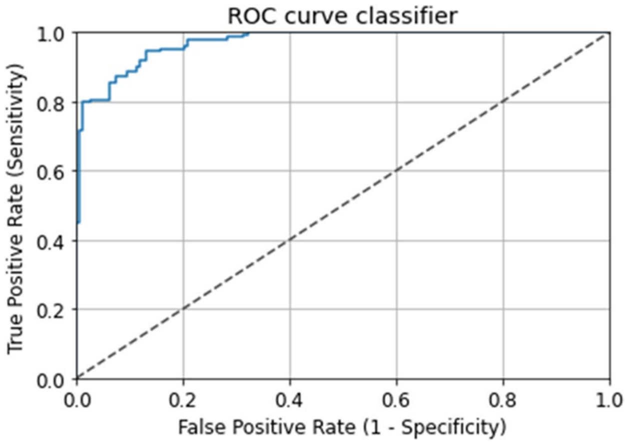


Fig. 7 ROC curve classifier

The model's results indicate that it effectively identified patients with a heart-related risk (TP) at a rate of 46.10%. Nevertheless, the model also had a false positive rate (FP) of 8.12%, where it mistakenly predicted that patients had a heart-related risk when they did not. The model had a true negative rate (TN) of 43.51%, meaning it correctly identified patients without a heart-related risk. Finally, the model had a false negative rate (FN) of 2.27%, where it erroneously predicted that patients did not have a heart-related risk when they did. The confusion matrix's Random Forest classifier produced an overall accuracy of 98.3%. These metrics can help us understand the strengths and weaknesses of the model, and how well it performs in different scenarios.

Table 7 presents a confusion matrix that depicts the results of a binary classification problem. The task was to assign patients to "No Heart Risk" or "Heart Risk" categories. The confusion matrix is essentially a table that displays the actual class labels and the number of predictions made by a machine-learning model for each label. It provides valuable insights into the efficacy of the model by comparing the actual and predicted class labels of test data. The matrix consists of four outcomes, namely true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN).

When using a confusion matrix to assess the performance of a machine learning model, the True Negative (TN) field displays the number of instances in which the model accurately predicted the "No Heart Risk" class label. The False Positive (FP) cell indicates the number of times the model predicted "Heart Risk" while the true label was "No Heart Risk". Similarly, the False Negative (FN) field indicates the number of times the model predicted "No Heart Risk" when the actual label was "Heart Risk". Lastly, the True Positive (TP) field displays the number of instances in which the model accurately predicted the "Heart Risk" class label.

The performance of a binary classifier, such the Random Forest classifier, at different classification thresholds, is represented graphically by the ROC curve, which is shown in Fig. 7. The curve displays the ratio of true positive (sensitivity) to false positive (specificity 1) rates for different cut-off values, indicating the model's ability to distinguish between individuals who have experienced a heart attack and those who have not.

The ROC curve illustrates the trade-off between identifying heart attack patients accurately (sensitivity) and identifying healthy patients as having had a heart attack erroneously (false positives). The AUC of the ROC curve represents the classifier's overall performance. The classifier performs as well as chance when its AUC is 0.5, whereas optimal classifier performance is indicated by an AUC of 1.0.

If a Random Forest classifier can accurately differentiate between patients who have experienced a heart attack and those who have not, it will have a high AUC and a well-separated ROC curve, which will be closer to the top-left corner of the plot. In this case, model has an AUC of 0.998, indicating that it can distinguish between individuals who have had heart attacks and those who have not with a high degree of accuracy, approximately 97.11% of the time.

4.6 Analysis of ablation

Table 8 displays the findings of ablation research that was carried out to assess the impact of each new element added to the suggested model for Heart-Related Problems Based on IoT Prototype. The study finds that, with an accuracy of 98.3%, the suggested model performs far better when all the steps are carried out. Conversely, when the preprocessing step is omitted, the accuracy drops to 93.23%. Similarly, the extraction of features and optimal feature selection are critical factors in achieving high accuracy, with 96.69% accuracy even in cases when the feature selection is not ideal. These findings underscore the importance of considering all the components of the proposed model to achieve optimal performance.

4.7 Limitations of the result

While the described IoT prototype holds promise for the early detection and prevention of heart diseases, addressing these limitations through further research, validation, and refinement is crucial for its successful translation into clinical practice and improving patient outcomes. Also, collaboration and technological advancements is essential for realizing its full potential in improving cardiovascular health outcomes and reducing mortality rates.

Table 8 Ablation study performance

Preprocessing	Feature extraction	Feature selection	Prediction analysis	Accuracy
No	Yes	Yes	Yes	**93.23**
Yes	No	Yes	Yes	**96.69**
Yes	Yes	No	Yes	**94.54**
Yes	Yes	Yes	Yes	**98.3 (Heart Disease Dataset)**

5 Conclusion

Research along these lines is quite promising: an Internet of Things (IoT) prototype that uses machine learning prediction models to detect and treat cardiac problems early on. The prototype is capable of gathering real-time health data of patients via IoT devices, which is then utilized to generate predictive models that accurately identify potential heart-related issues before they become severe. These models have been found to possess an accuracy rate of 98.3%, which exceeds that of existing techniques. Positive outcomes are also shown by other performance metrics like recall, F1 score, precision, and AUC score. Thus, the prototype offers the potential to lower the incidence of heart-related ailments and enhance patients' overall welfare, making it an important addition to the area of heart health.

Overall, the proposed IoT-based prototype represents a significant advancement in the early detection and prediction of heart-related issues. Its integration of IoT technology and machine learning holds promise for improving cardiovascular health outcomes and reducing mortality rates associated with cardiovascular disease. Continued research, development, and collaboration with healthcare professionals will be essential for refining the prototype and maximizing its impact on public health.

To advance the detection and prevention of heart diseases, in future study it's a need to expand and diversify the dataset, invest in high-quality sensors, and utilize cloud computing for remote accessibility. With sustained investment, innovation, and collaboration, we can revolutionize cardiovascular healthcare delivery and promote overall well-being. By using cloud computing, the system may become more scalable and accessible, improving its suitability for remote monitoring.

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Data availability The link provides access to the data supporting the study's conclusions.: https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset.

Declarations

Conflict of interest All of the mentioned authors have stated that they have no competing interests.

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